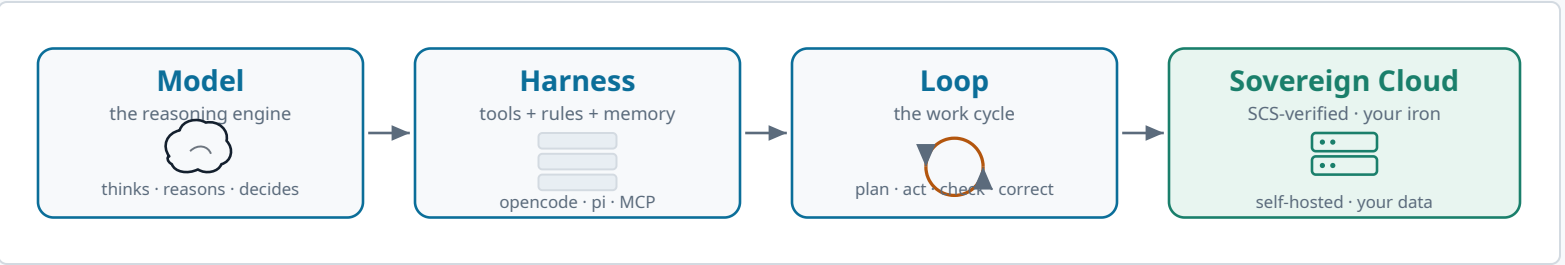
SOVEREIGN CLOUD DAYS 2026 · WORKSHOP STAGE

Agentic AI meets Sovereign Cloud

openEduSuite on SCS-verified legacy bare-metal hardware

Sun, 25.10.2026 · 11:15 · 60 min incl. Q&A

Tobias Weiß · HRZ Philipps-Universität Marburg



MOTIVATION · 3 MIN

Legacy hardware isn't junk

- Data centers have **decommissioned servers** — often fully functional.
- A sovereign, AI-capable setup runs on **written-off iron**.
- 3 × Huawei RH2288E V2 (~2013) → SCS-verified K3s cluster.

> openEduSuite: HFD Digital Sovereignty Ideas Competition — winner 2026-10-06.

WS **Who runs “legacy hardware” in production themselves?**

MOTIVATION · THE PIVOT · 3 MIN

From OpenStack to bare-metal K3s

OpenStack / OVN on VMs

OVN 25.03.1 gateway bug
FIPs dead, DHCP broke

“Reboot destroys the fix”

Pivot

Bare-metal K3s (SCS-verified)

3 nodes · Flannel · MetalLB · Ceph
6/6 KaaS standards PASS · 444/444 CNCF

fewer layers, fewer failure points

What broke in OVN 25.03.1: gateway redirect bug · floating IPs (FIPs) unreachable · DHCP leases lost · OVN northd/southbound DB + OVS bridges on every node · “reboot destroys the fix” — every node restart re-triggered it.

The VM/SDN abstraction was too heavy for 3 nodes — bare metal is leaner.

HANDS-ON · INSTALL · 2 MIN

Install your tool — pick one

Beginners · opencode

```
curl -fsSL https://opencode.ai/install | bash
# Or: npm install -g opencode-ai
# Or: bun add -g opencode-ai (much faster)
# Verify: opencode --version
# Start: cd <project> && opencode
```

Advanced · pi.dev

```
curl -fsSL https://pi.dev/install.sh | sh
# Or: npm i -g @earendil-works/pi-coding-agent (Node 22.19+)
# Or: bun add -g @earendil-works/pi-coding-agent (much faster)
# Verify: pi --version
# Start: cd <project> && pi
```

Both are free to start — no paid API key required. Using **bun** gives a noticeable speedup.

HANDS-ON · CHOOSING A MODEL · 3 MIN

Start for free — Big Pickle or Beam

- **opencode (beginners):** `/connect` → **OpenCode Zen** → pick **Big Pickle** or **Beam Free**
- **pi.dev (advanced):** `/login` → provider **opencode**, then `/model` → `big-pickle` or `beam-free`
- Alternatives: any provider via `/connect` (75+) or your own API key

> Both are free on OpenCode Zen — enough for agentic experiments, no cost risk.

> ⚠️ **Privacy — read before you paste.** Free cloud models differ in how they handle your data. **Big Pickle** is a stealth model; its provider **may use your input to train**. **Beam Free** (Reflection's open-weight 501B model, 1M-token context) does **not** train on your data (retains it 30 days for safety). **Never paste private, customer, or unpublished data into a model that trains on user input.** For sensitive work, run a local model (Ollama, `ollama run beam`) or your own sovereign endpoint.

HANDS-ON · PROMPT IDEAS · 6 MIN

Starter prompts to try

- “Explain SCS-0217 Pod-Security in 3 sentences.”
- “Write a default-deny NetworkPolicy for the `llm` namespace.”
- “PVC on ceph-rbd, 3× replication — show the manifest.”
- “Ceph is HEALTH_WARN — give 3 first debugging steps.”
- “Outline: student question → retrieval → Ollama CPU → answer.”

WS **Who has opencode running in 5 min? → walk around, help.**

CLUSTER · HARDWARE · 3 MIN

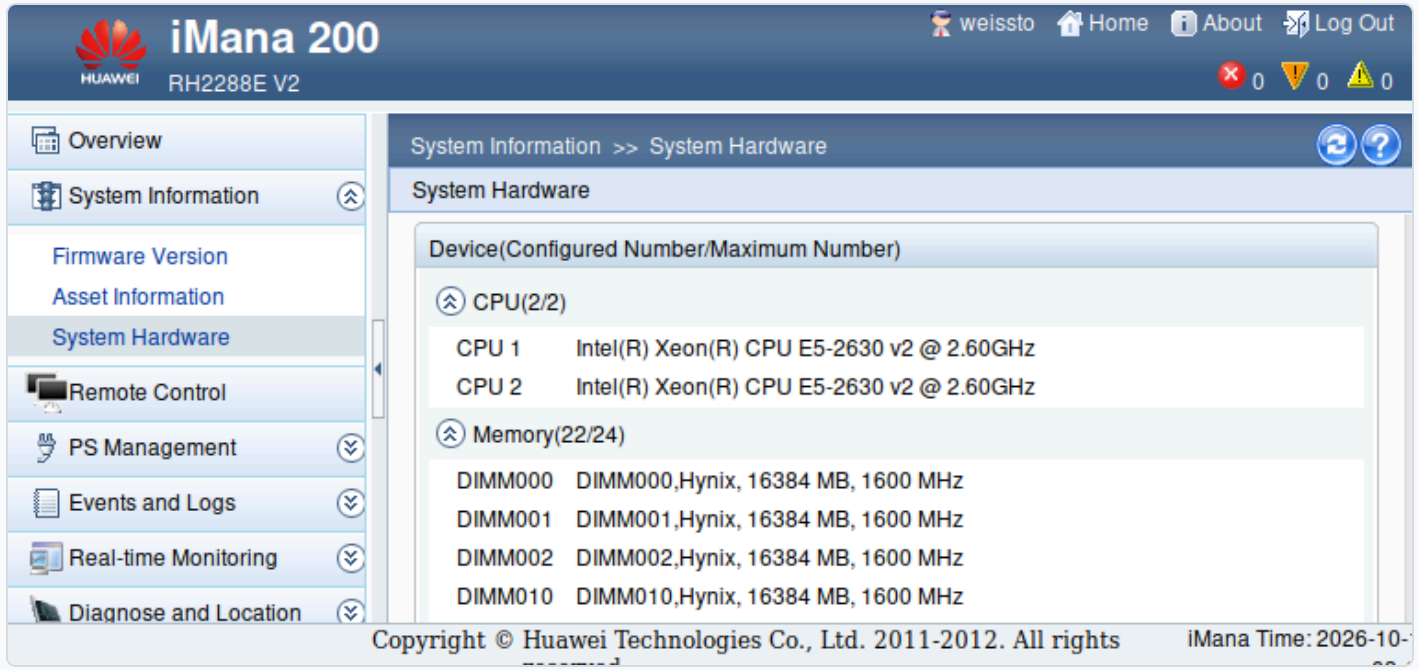
3 × Huawei RH2288E V2 — no GPU

CPU
2× Xeon E5-2600 v2 (Ivy Bridge-EP) · 6–12 cores

RAM
24× DDR3 ECC RDIMM · max 768 GB

STORAGE
6× 512 GB SSD · Ceph 8 OSDs · Repl. 3

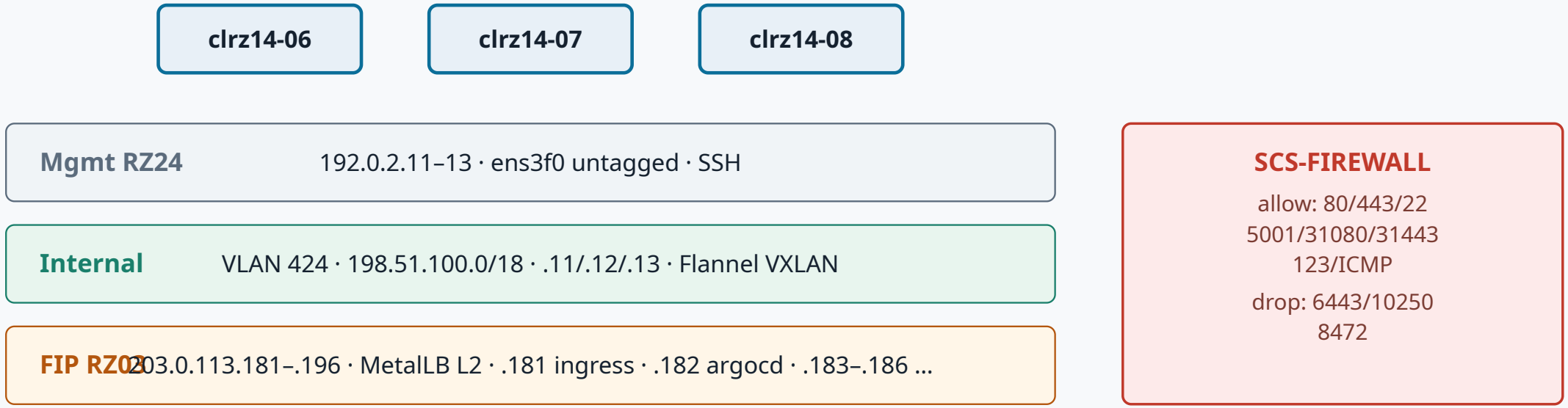
NETWORK
2× Intel X540-AT2 10 GbE



iBMC iMana — hardware view (clrz14-06/07/08)

CLUSTER · NETWORK · 3 MIN

Three networks, one firewall



ens3f0: mgmt + FIP (VLAN 3) · ens6f0: internal (VLAN 424) · iBMC out-of-band

CLUSTER · K3S STACK · 4 MIN

The stack — bare-metal in production

openEduSuite Workloads (edu · staff · students · sme · llm)

ArgoCD (GitOps) · K8up/Restic (backup → SeaweedFS S3)

Ceph 18.2.8 Reef — 8 OSDs, RBD + CephFS, Repl. 3

Flannel VXLAN · MetalLB L2 · HAProxy Ingress

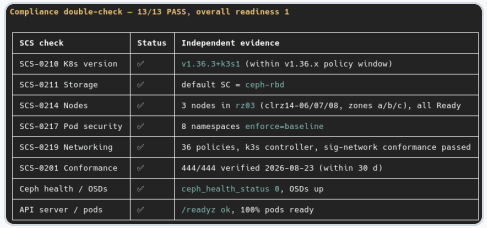
K3s v1.36.3+k3s1 · Single-server · embedded datastore

Ubuntu 24.04 LTS · containerd 2.3.3 · runc 1.4.3

3 × Huawei RH2288E V2 (bare metal)

CLUSTER · SCS-VERIFIED · 3 MIN

6/6 KaaS standards PASS — measurable



Compliance double-check — 13/13 PASS, overall readiness 1

SCS check	Status	Independent evidence
SCS-0210 K8s version	✓	v1.36.3+k3s1 (within v1.36.x policy window)
SCS-0211 Storage	✓	default SC = ceph-rbd
SCS-0214 Nodes	✓	3 nodes in rz03 (clrz14-06/07/08, zones a/b/c), all Ready
SCS-0217 Pod security	✓	8 namespaces enforce=baseline
SCS-0219 Networking	✓	36 policies, k3s controller, sig-network conformance passed
SCS-0201 Conformance	✓	444/444 verified 2026-08-23 (within 30 d)
Ceph health / OSDs	✓	ceph_health_status 0, OSDs up
API server / pods	✓	/readyz ok, 100% pods ready

SCS KaaS compliance check (2026-08-23)

STANDARD	TEST	RESULT
SCS-0201 CNCF conformance	`cncf-k8s-conformance` (444/444 specs)	✅ PASS
SCS-0210 version policy	`version-policy-check` (v1.36.3 in window)	✅ PASS
SCS-0211 default storage class	manual (ceph-rbd, repl. 3)	✅ PASS
SCS-0214 node distribution	manual (3 nodes, zones a/b/c)	✅ PASS
SCS-0217 pod security	manual (8 NS baseline; 2 system NS privileged)	✅ PASS
SCS-0219 KaaS networking	`kaas-networking-check` (37 NetworkPolicies)	✅ PASS

Open: OIDC as 2nd auth · Pod-Security on more NS · Loki/Trivy CI · Ceph at rest · GDPR clearance.

SELF-CHECK · EU CLOUD SOVEREIGNTY SCORE · 2 MIN

73.5 % Sovereignty Score — SEAL-1

Method: **EU Cloud Sovereignty Framework** (official calculator, Jun 2026) — weighted **Sovereignty Score** + **SEAL** level (weakest link across all 48 criteria).

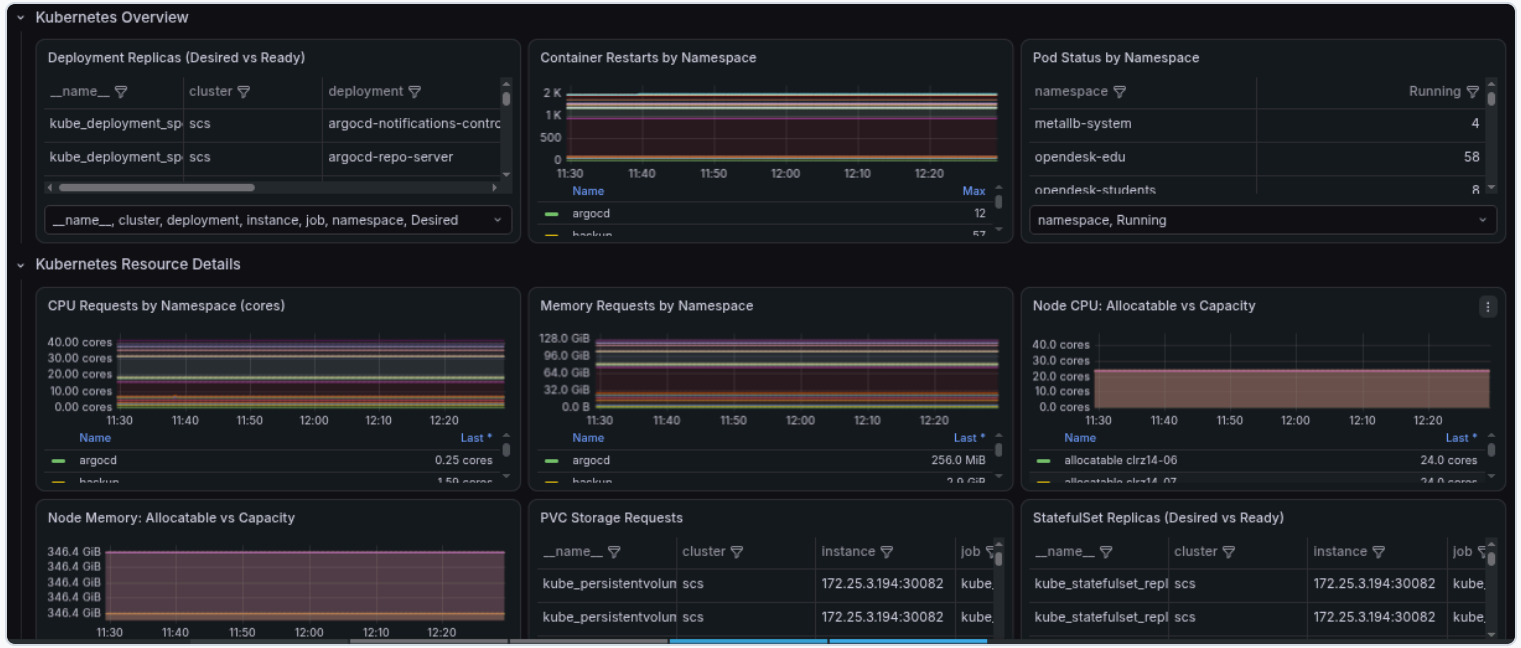
OBJECTIVE	W	SCORE	SEAL
SOV-1 Strategic	20 %	**94 %**	SEAL-4
SOV-4 Operational	15 %	**83 %**	SEAL-3
SOV-6 Technology	15 %	**80 %**	SEAL-3
SOV-3 Data & AI	10 %	75 %	SEAL-1
SOV-2 Legal & Jurisdictional	10 %	71 %	SEAL-1
SOV-7 Security & Compliance	15 %	64 %	SEAL-1
SOV-5 Supply Chain	10 %	50 %	SEAL-1
SOV-8 Environmental	5 %	19 %	SEAL-1

High score — low SEAL. SEAL is the **weakest link**: non-EU hardware (Huawei/Intel, SOV-5), no formal EAL certification (SOV-7), no PUE/emissions reporting (SOV-8).

Levers to SEAL ≥ 3: EU-fabricated platform / hardware provenance · formal EAL cert · PUE & CSRD reporting.

CLUSTER · OPERATIONS · 2 MIN

What we learned in operation



Grafana K3s dashboard (7 dashboards, predictive-agent)

- Stale digests, default-deny NetworkPolicies, OIDC migration — all resolved and documented
- Updates: Ceph-first, rolling, rollback script · backup: K8up → SeaweedFS S3

OPENEDUSUITE · WORKLOAD · 4 MIN

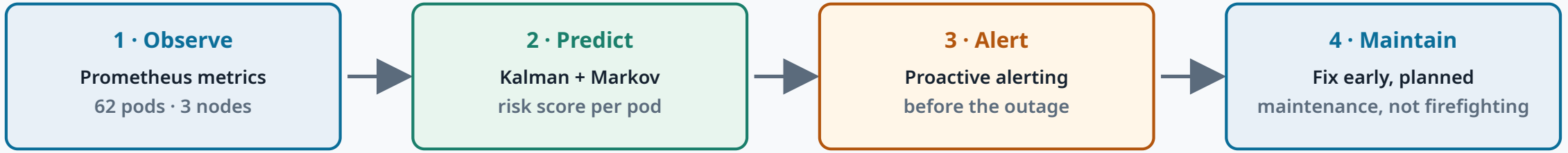
The dimensions of openEduSuite

DIMENSION	COMPONENTS
Tenants — who	edu · staff · students · sme · llm · predictive-agent
Identity & access	Keycloak / OIDC · OpenLDAP (.186)
Communication	Postfix (.185) · Stalwart (.183) · DKIMpy (.184) · Matrix
Collaboration	Nextcloud · XWiki · SOGo6 · OpenProject
Delivery	Own Nix builds on GHCR → local Zot registry

> **Honest status:** `opendesk-sme` still Degraded (stale digests).

OPENEDUSUITE · AGENTIC WORKFLOW · 4 MIN

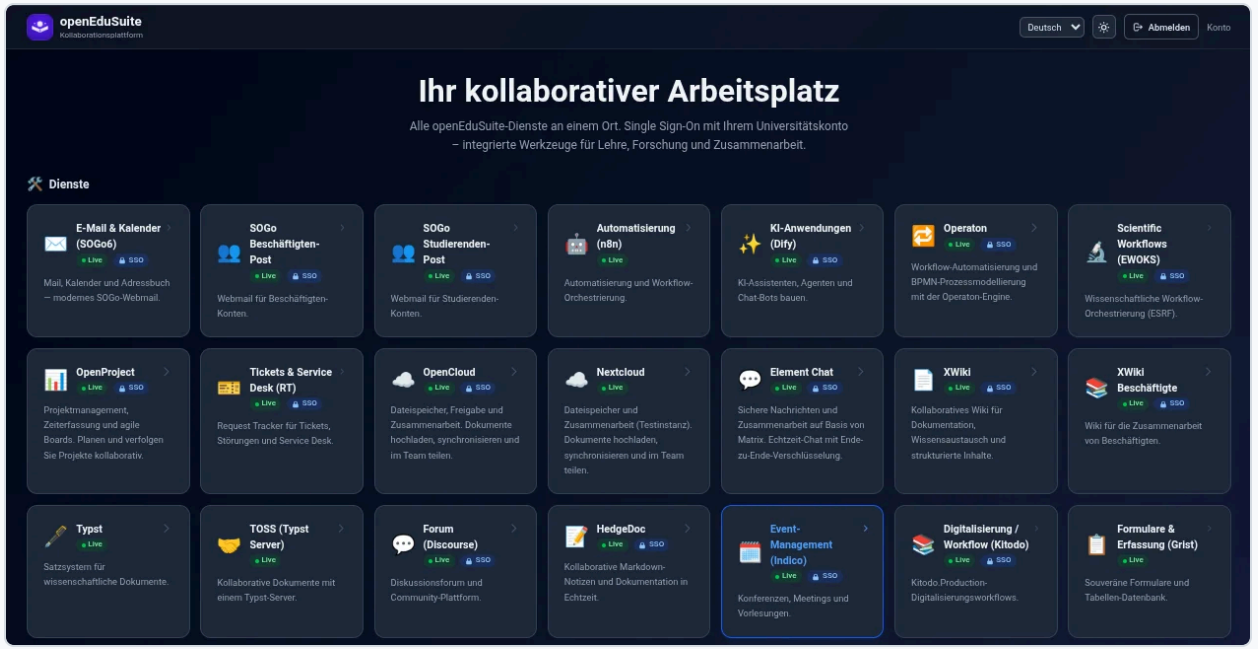
Cluster maintenance in 4 steps



predictive-agent v8.11-nix (own tenant): Risk scoring (Kalman + Markov) for 62 pods — alerts before users notice.

OPENEDUSUITE · LIVE DEMO · 4 MIN

Cluster, login, agent trace



openEduSuite — bird's-eye view of the platform

- Dashboard: 3 nodes · Ceph HEALTH OK (8/8) · SCS board
- Login via Keycloak/OIDC → agentic workflow

> **Fallback:** Screenshots (Grafana, KVM, open-webui) + stack diagram if the demo stalls.

LESSONS · 3 MIN

Five takeaways

- **Start small, learn fast** — OpenStack didn't fit the scale.
- **Bare metal instead of a VM layer** — less abstraction, more control.
- **Constraints are features** — quantization beats buying GPUs.
- **“Verified” convinces** — SCS tests open doors, with honest gap reporting.
- **Document like a madman** — runbooks, DPIAs, checklists.

Q&A · 10 MIN

Questions & discussion

- Architecture, rebuilding it, SCS/openEduSuite licensing topics
- Workshop element: opencode install + the prompts — try them at home
- Contact: tobias-weiss.org · graphwiz.ai · ki-kompetenz-training.org

> **Backup slides:** architecture diagram · agent trace · cluster fact sheet.

BACKUP · ARCHITECTURE

Cluster architecture — full picture

clrz14-06 — master
K3s server (single) · Ceph MON · 4 OSDs
mgmt 192.0.2.11 · int .11
2× Xeon E5-2600 v2 · no GPU

clrz14-07 — worker
K3s agent · Ceph MON · 2 OSDs
mgmt 192.0.2.12 · int .12
openEduSuite + llm workloads

clrz14-08 — worker
K3s agent · Ceph MON · 2 OSDs
mgmt 192.0.2.13 · int .13
predictive-agent · monitoring

K3s v1.36.3+k3s1 · Flannel VXLAN 10.42.0.0/16 · MetalLB L2 · HAProxy Ingress · ArgoCD v2.12.6

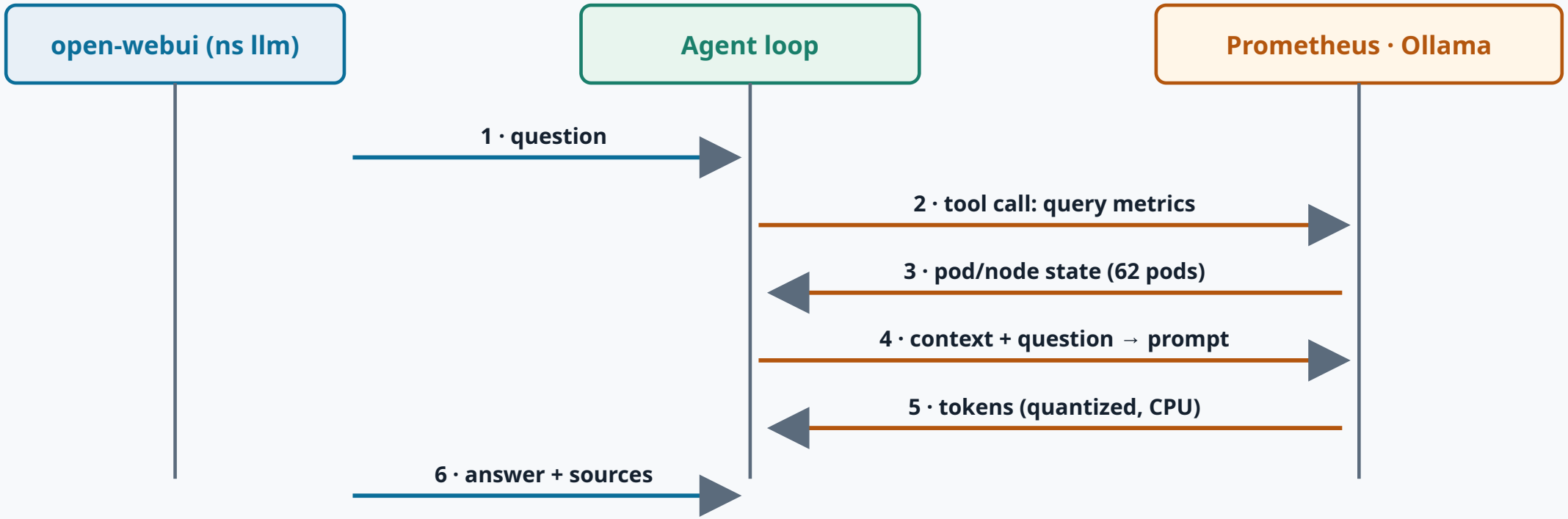
Ceph 18.2.8 Reef — 8 OSDs · 3 MONs · repl. 3 · RBD + CephFS · K8up/Restic → SeaweedFS S3

Ubuntu 24.04 LTS · containerd 2.3.3 · runc 1.4.3 · firewall SCS-FIREWALL on ens3f0

Mgmt	192.0.2.11–13 · ens3f0 untagged · SSH
Internal	VLAN 424 · 198.51.100.0/18 · .11/.12/.13
FIP	203.0.113.181–.196 · .181 ingress · .182 argocd · .183–.186 mail/ldap

BACKUP · AGENT TRACE

How an agent call traces through the cluster



Multi-step, tool-using, verifiable — not a single prompt. Same pattern as the predictive-agent risk loop.

BACKUP · CLUSTER FACT SHEET

openEduSuite cluster — fact sheet

HARDWARE

3 × Huawei RH2288E V2 · 2 × Xeon E5-2600 v2 · no GPU · 6 × 512 GB SSD each · 2 × Intel X540-AT2 10 GbE **[verify]**

STORAGE

Ceph 18.2.8 Reef · 8 OSDs / 3 MONs · repl. 3 · RBD + CephFS · K8up/Restic → SeaweedFS S3

WORKLOADS

openEduSuite (edu · staff · students · sme) · llm (open-webui, Ollama) · predictive-agent v8.11-nix · 62 pods

PLATFORM

K3s v1.36.3+k3s1 (single server) · Flannel VXLAN · MetalLB L2 · HAProxy Ingress · ArgoCD v2.12.6

SCS COMPLIANCE

6/6 KaaS standards PASS · 444/444 CNCF · DevSecOps 78/100 (C+) · ~40 NetworkPolicies

OS / OPS

Ubuntu 24.04 LTS · containerd 2.3.3 · Prometheus + Grafana (7 dashboards) · firewall SCS-FIREWALL

Nix-built images on GHCR → local Zot registry · opendesk-sme still Degraded (stale digests) · DPIA/TOM/ROPA documented, personal data approval pending.